Retained Hedging Models: Surface-Dynamics Audit

AI-agent implementation and diagnostic feedback

14 November 2026

Executive update: 14 November 2026

Standalone DYSANOS v2 and the earlier STRT-16 alternative are distinct models. Blue values throughout this document are earlier, unrefreshed evidence, not current refits. Current black/orange/red ratings retain their stated diagnostic meanings. Blue overrides the earlier traffic-light colors in the historical appendix, including plot numbers.

Table 1: Model-fit counts out of 44 April–May 2024 origins. Training does not imply forecast or diagnostic completion. Black: ungraded counts; no missing result is a pass. Color rules: Black: ungraded entries. No orange or red grades are assigned. Missing graded entries are orange; other raw quantities are ungraded.

Model	Completed origins	Current stage
DYSANOS v1-P	1	paper fits; scoring pending
Standalone DYSANOS v2-P	44	44/44 one-day forecasts
STRT-16 alternative	44	earlier separate experiment
GROMM2-P/Q	1	fits; P/Q forecasts pending

1 Standalone DYSANOS v2: matched-date results

This is the 41-expiry density-weighted standalone model, with sixteen factors, a 250-change age-weighted window and Gaussian joint spot/factor innovations. It is distinct from the older nine-expiry STRT-16 alternative. The input SANOS smoothness is 0.1; the common observable reference has smoothness 0.25.

Table 2: Standalone V2 same-day reconstruction and out-of-sample straddle sensitivity-span capture. Each realized next-day change is projected ex post onto model sensitivities; these are not forecast variance or attainable trading returns. Black entries are descriptive and ungraded. Color rules: Black: ungraded entries. No orange or red grades are assigned. Missing graded entries are orange; other raw quantities are ungraded.

Dates	Quotes	Inside spread %	Spot + 8 %	Spot + 16 %
44	103686	85.891	94.095	96.752

Table 3: Standalone V2 one-day Gaussian forecasts, 1,000 paths per origin, in the common 49-node observable grid. Conditional and pooled traces use observed change covariance as denominator; variance/MSE uses squared forecast errors. Black entries are descriptive and ungraded. Color rules: Black: ungraded entries. No orange or red grades are assigned. Missing graded entries are orange; other raw quantities are ungraded.

Dates	Cond. trace	Pooled trace	Frob.	RMSE	Variance/MSE
44	0.238	0.585	0.440	0.171	0.302

Table 4: Pooled spot–surface correlations in the fixed observable PCA basis. Black entries are descriptive and ungraded. Model and realized spot returns use the same forward numeraire. Color rules: Black: ungraded entries. No orange or red grades are assigned. Missing graded entries are orange; other raw quantities are ungraded.

Observable PC	Model spot corr.	Data spot corr.
1	-0.449	-0.680
2	0.308	0.157
3	0.137	-0.008

Table 5: Standalone V2 Gaussian one-day PC1 tail diagnostics. Forty-four dates are too few for conclusive 1% or 5% tail calibration. Color rules: PC1 exceptions: red for exact-binomial p below 0.01; orange for p below 0.05 or fewer than 10 expected exceptions; black otherwise. Other entries are ungraded. Missing graded entries are orange; other raw quantities are ungraded.

Quantile	Dates	PC1 exceptions	Expected	Exact p	Pinball
0.01	44	4	0.440	0.000987	0.059
0.05	44	5	2.200	0.067	0.137
0.95	44	9	2.200	0.000280	0.167
0.99	44	4	0.440	0.000987	0.068

The maximum fitted VAR spectral radius is 0.9690; this checks the fitted factor backbone, not the full nonlinear surface process.

Forecast changes are measured relative to each model's own initial surface. Standalone V2 spot shocks are transformed deterministically from the fitted dividend-adjusted return to the common one-day-forward convention. Long-horizon recursive stability and static-arbitrage simulation remain pending for this standalone model.

2 Earlier STRT-16 alternative and competitors

The STRT-16 alternative predicts 0.508 times realized squared forecast-error variance. Its common log-IV forecast RMSE is 0.1187, versus 0.1904 for persistence. All 44 fitted floating VAR backbones are stable, with spectral radii 0.9253–0.9550. No tested static-arbitrage violation or unusable IV occurs in these checks. These are guarded-decoder results, not unguarded diffusion or dynamic-martingale proofs.

All rows use the same 44 forecast dates, common 49-coordinate floating log-IV grid and pre-April diagnostic PCA. Let $m_t = E_t[\Delta x]$ and $C_t = \mathrm{Cov}_t(\Delta x)$. Conditional trace divides $\mathrm{tr}(\bar{C})$ by observed change covariance trace; pooled trace adds the across-origin covariance of m_t. These are different from forecast variance/MSE, $\sum_t \mathrm{tr}\, C_t / \sum_t \|\Delta x_t - m_t\|^2$. Smaller pooled Frobenius error and forecast RMSE are better; RMSE/persistence below one beats the zero-change forecast. None is an arbitrage test.

Table 6: Earlier STRT-16 and competitor results, not the standalone DYSANOS v2. Blue entries are retained earlier evidence, not current refits. Color rules: Black: ungraded entries. No orange or red grades are assigned. Missing graded entries are orange; other raw quantities are ungraded.

Model	Dates	Cond. trace	Pooled trace	Frob. pooled	RMSE	/persist.
STRT-16 alternative	44	0.193	0.671	0.385	0.119	0.624
FuNVol	44	0.397	0.495	0.580	0.183	0.964
VolGAN PCA8	44	0.000251	0.970	0.283	0.215	1.127
AD-Seq-Vol	44	0.374	1.542	0.786	0.237	1.244
AD-Seq-Vol-FT	44	0.302	1.455	0.697	0.266	1.396

STRT-16 is an earlier spot-orthogonal tangent projection followed by a separately fitted daily VAR and scalar tail diffusion. Native spot loading outside the retained span is preserved. All fits use independent causal windows, without earlier-origin weights. Decoder guards activate frequently, so static validity does not establish calibrated diffusion tails.

2.1 Current reconstruction and next-day price forecast

Table 7: Earlier STRT-16 on the matched diagnostic book: OTM, 1¡business-days-to-expiry¡=5 and absolute NMS¡=1.5. Statistics are quote-count weighted across dates. Outside distance uses full spread width. Next-day point forecasts versus realized quotes are distinct from same-day reconstruction. Color rules: Black: ungraded entries. No orange or red grades are assigned. Missing graded entries are orange; other raw quantities are ungraded.

Mode	Quotes	Inside spread %	Mean outside	MAE / spot
reconstruction	8358	93.587	0.081	0.000009
next day	7610	1.051	36.153	0.002

2.2 Static arbitrage and simulated-IV stability

Each of 44 origins uses 1,000 paths, nested simulation days 1, 5, 10, 125 and 259, twenty-one NMS strikes in [-3,2], and remaining tenors 1, 2, 5 and 10 business days. Calendar checks use the next tenor in the native model grid at the same cash strikes. No separate stock path is needed. Prices and vector IV inversion use the existing decoder and cdxcore.bs without rejected-path filtering.

Table 8: Earlier STRT-16 violation counts summed across all 44 origins at each horizon and tenor. Zero violations in finite paths do not prove absence of arbitrage. Color rules: Blue: retained historical counts; no current grading. Missing graded entries are orange; other raw quantities are ungraded.

Day	Tenor	Decr.	Convex	Vertical	Intrinsic	Upper	K=0	Calendar	Bad IV
1	1	0	0	0	0	0	0	0	0
1	2	0	0	0	0	0	0	0	0
1	5	0	0	0	0	0	0	0	0
1	10	0	0	0	0	0	0	0	0
5	1	0	0	0	0	0	0	0	0
5	2	0	0	0	0	0	0	0	0
5	5	0	0	0	0	0	0	0	0
5	10	0	0	0	0	0	0	0	0
10	1	0	0	0	0	0	0	0	0
10	2	0	0	0	0	0	0	0	0
10	5	0	0	0	0	0	0	0	0
10	10	0	0	0	0	0	0	0	0
125	1	0	0	0	0	0	0	0	0
125	2	0	0	0	0	0	0	0	0
125	5	0	0	0	0	0	0	0	0
125	10	0	0	0	0	0	0	0	0
259	1	0	0	0	0	0	0	0	0
259	2	0	0	0	0	0	0	0	0
259	5	0	0	0	0	0	0	0	0
259	10	0	0	0	0	0	0	0	0

The largest decoded annual IV is 38.16%; no IV exceeds 300%. No inversion is unusable. Numerical DLV/forward bounds remain active and are disclosed in the dated records. Historical-range exceedances are recorded, not interpreted as arbitrage.

2.3 Frozen observable-PC persistence and tails

Table 9: Mean within-path Pearson ACF: first 32 of the same 1,000-path stream, 115 sessions, all 44 origins. Historical columns average each origin's causal training window, not a 44-day held-out ACF. A fixed pre-April observable basis is used throughout. Black: ungraded descriptive comparisons; no color inference is assigned. Color rules: Black: ungraded entries. No orange or red grades are assigned. Missing graded entries are orange; other raw quantities are ungraded.

Lag	Model PC1	PC2	PC3	History PC1	PC2	PC3
1	0.764	0.738	0.624	0.791	0.804	0.661
5	0.379	0.426	0.216	0.428	0.435	0.210
21	0.000099	0.007	0.009	-0.006	0.033	-0.010
63	0.006	0.003	-0.003	-0.336	-0.318	-0.185

Table 10: Earlier STRT-16 PC1 pointwise tail diagnostics. With only 44 dates all tails remain exploratory. Color rules: Blue: retained historical statistics; no current grading. Missing graded entries are orange; other raw quantities are ungraded.

Quantile	Dates	PC1 exceptions	Expected	Exact p	Pinball
0.01	44	2	0.440	0.072	0.017
0.05	44	6	2.200	0.022	0.084
0.95	44	6	2.200	0.022	0.111
0.99	44	3	0.440	0.010	0.038

2.4 Outstanding work and objective differences

DYSANOS v1 uses the paper's full joint affine-200/five-session VAR. The replacement run centers its quote loss on observed bid/ask midpoints; the earlier SANOS-centered trial is not reused as a paper-objective fit. An intermediate, unaccepted weekly covariance from that earlier trial failed the daily-embedding positivity check; no eigenvalue clipping is used. No completed 44-origin result is claimed. GROMM2 uses a separately identified calibration with actual observed bid/ask, the paper's linear excess and 0.05 midpoint term plus restoring cap penalty. The old pseudo-Huber chunks are not loaded. Its requested GROMM2 geometry and latent-X dependence estimator remain explicit extensions. GPU training and GPU-dependent forecasts share one managed allocation, released when that work ends; remaining statistics run on CPU.

A Historical evidence: blue values are not updated refits

The following earlier report is retained for traceability. Its numbers, former quality colors and plot labels are blue. Its completion claims refer to those older model variants only, not to the currently requested DYSANOS v1/v2 and repository GROMM2 refits. Refer to the current coverage table for outstanding work.

Executive summary

Purpose. Assess the surface dynamics of retained rolling models on 44 April–May 2024 forecast origins, using one common log-IV grid and a PCA basis fixed before April. This is a model-law audit, not a rerun of hedging performance or a proof of no arbitrage.

- **DYSANOS: stationary backbone, weak observable dispersion.** All 88/88 distinct saved floating-volatility PCA–AR(1) backbones are stationary (maximum spectral radius 0.735). Nevertheless, pooled modeled log-IV-change variance is only 16.6% of observed variance for ordinary PCA and 21.4% for spot-orthogonal PCA. Stationarity is not the same as calibration.
- **VolGAN: aggregate agreement hides conditional collapse.** Pooled modeled variance is 97.0% of observed variance, but average within-origin forecast variance is only 0.025%. A pooled variance ratio near one can therefore be misleading.
- **Judge more than variance scale.** Compare covariance shape, spot leverage, forecast skill relative to persistence, and recursive behavior. A trace ratio near one is not an overall pass, and the conditional ratio need not equal one because its observed denominator is unconditional. GROMM-Q comparisons with historical P outcomes are descriptive, not P-calibration requirements.
- **Tail conclusions remain preliminary.** Forty-four realized dates imply only 0.44 expected exceptions at a 1% tail and 2.2 at a 5% tail. Non-rejection cannot establish tail calibration. Stable AR coefficients and finite simulated paths do not prove non-explosion of the complete decoded model.
- **Coverage at this report build.** Completed model–origin checks: surface forecasts 440/440; quote forecasts 440/440; recursive path tests 440/440. All requested origin-level diagnostics are complete; completion does not imply a model-quality pass.

Interpretation. These are diagnostics of retained implementations, including disclosed numerical guards and differing SANOS input smoothnesses, not a controlled architecture-only comparison. Black/orange/red colors use the explicit review tolerances in each table caption; they are not universal statistical acceptance criteria.

Completed density-weighted follow-up. Both additional sixteen-factor DYSANOS variants have all 44 surface, quote and recursive diagnostics completed. Pooled log-IV variance ratios improve to 0.784 and 0.722; weighted forecast accuracy is close to GROMM. However, predicted variance is only about 0.45–0.51 times realized squared forecast errors, versus 1.82–2.06 for GROMM; a larger conditional variance bar is therefore not automatically better calibrated. All weighted recursive paths decoded without invalid or above-300-percent-IV surfaces, but native clipping persists, tail/ACF mismatches remain, and next-day quoted-price MAE is slightly worse than persistence. The spot projection/add-back and all covariance cross terms were checked explicitly. See the completed follow-up section for the full definitions and limitations.

B Scope and headline findings

The audit covers 44 origins from 1 April through 31 May 2024; the final one-session forecast settles on 3 June. Existing fits are retained unchanged. DYSANOS means our own saved floating-DLV PCA/AR(1) fits, not the latest external SANOS package. Results are diagnostics of retained legacy fits, not validation of the corrected hedge2 specification.

The observable reference is the retained public-candidate SANOS interpolation (smoothness 0.25), manifest `2add65d6...`. Our DYSANOS coordinates instead came from the licensed smoothness-0.1 interpolation. These are not identical input pipelines; the comparison therefore measures retained-system behaviour, not architecture alone.

Main distinction: pooled covariance can look realistic even when a model has almost no conditional randomness. VolGAN is the clearest example. Persistence is measured using one pre-April observable PCA basis, never by concatenating rolling native factor scores. The DYSANOS stationarity assessment concerns only its floating-volatility PCA–AR(1) backbone; spot-level stationarity is neither required nor tested. All 44 fitted backbones in each DYSANOS variant admit a stationary distribution.

Table 11: Diagnostic coverage: dates evaluated out of the 44 requested April–May 2024 origins. Color rules: Black: all 44 requested origins evaluated. Orange: fewer than 44. No red coverage grade; coverage is not model quality. Missing graded entries are orange; other raw quantities are ungraded.

Model	Requested dates	Surface dates	Quote dates	Path-test dates
DYSANOS ordinary (16 factors)	44	44	44	44
DYSANOS spot-orthogonal (16 factors)	44	44	44	44
FuNVol	44	44	44	44
VolGAN PCA8	44	44	44	44
AD-Seq-Vol	44	44	44	44
AD-Seq-Vol-FT	44	44	44	44
GROMM-P	44	44	44	44
GROMM-Q	44	44	44	44
DYSANOS ordinary (16 factors) (tail)	44	44	44	44
DYSANOS spot-orthogonal (16 factors) (tail)	44	44	44	44

Table 11 is a coverage inventory, not a performance or pass/fail table. Surface dates count origins with evaluated next-day IV forecasts for covariance and tail diagnostics. Quote dates count origins with exact next-day contract matches for price-forecast scoring. Path-test dates count completed recursive floating-IV tests, including tests that produced no wholly valid decoded IV paths. Such failures contribute their actual invalid rates to Table 25; they are not hidden as missing coverage. Each entry counts dates, not Monte Carlo paths; zero means no completed test, not a model-quality failure.

Covariance/tail availability follows retained next-surface arrays or newly sampled saved fits. The base-diffusion first next-IV artifact was absent: 512 new CPU scenarios were generated from its existing

1 April EMA checkpoint, with no retraining; its other 43 dates retain their 10,000-path forecasts. At the user's request, 22 additional May FT fits now continue the original April chain sequentially, using the unchanged selected one-epoch rolling update and causal 252-observation windows. Training completion is distinct from forecast-scenario availability in Table 11. Quote forecasts can use retained price-return books independently of IV arrays. The previously missing 20 May quote join was a normalized-strike float32 round-trip identity bug, not absent quotes. The corrected join accepts a source cash strike only when it is the unique candidate within four float32 machine epsilons times the reconstructed cash-strike scale. Ambiguous and more distant contracts remain excluded; per-date maximum identity errors are retained.

Every model uses the same 32-path, 115-business-session protocol on the same 44 origins, with seed 20261111. Completed neural paths reuse their identical first 115 sessions; partial diffusion checkpoints preserve the original context and RNG state. Native models retain their different stochastic laws, not artificially identical generated prices. GROMM's final saved expiry is 125D and the observable grid extends to 10D, so 115 sessions is the common horizon without inventing expiries or long-end extrapolation. Its aged maturities are decoded by total-variance interpolation at fixed normalized strike, with a zero-maturity variance anchor (flat short-end IV). All models use their own generated history, never future observed states. Coverage counts only completed tests; queued work is not a pass.

C Traffic-light interpretation and origins of thresholds

Black means within the stated review tolerance, orange means warning or insufficient evidence, and red means outside tolerance or a rejected diagnostic null. These are diagnostic-specific labels, not a proof of arbitrage-freedom and not a ranking of trading profitability. Counts, reference values and proper scores without a comparison baseline are shown neutrally. Missing evidence is orange, never a pass.

Table 12: Retrospective traffic-light rules. Black: pass under the stated review tolerance; orange: warning or insufficient evidence; red: outside tolerance or a rejected diagnostic null. Missing evidence is orange. Boundaries are inclusive except where stated.

Diagnostic	Pass (black)	Warn (orange)	Fail (red)
Covariance Frobenius error	at most 0.25	above 0.25, at most 0.50	above 0.50
Variance trace ratio	0.80 to 1.25	0.50 to 2.00, outside pass	outside 0.50 to 2.00
Leading-share L1 error	at most 0.10	above 0.10, at most 0.25	above 0.25
Subspace angle	at most 15 degrees	above 15, at most 30 degrees	above 30 degrees
Spot-correlation or IV-ACF error	at most 0.10	above 0.10, at most 0.20	above 0.20
Forecast loss / persistence	at most 1.00	above 1.00, at most 1.10	above 1.10
Reconstruction inside spread	at least 95%	80% to below 95%	below 80%
Reconstruction mean outside distance	at most 0.25 spreads	above 0.25, at most 1 spread	above 1 spread
Reconstruction q95 outside distance	at most 1 spread	above 1, at most 2 spreads	above 2 spreads
May/April relative-skill deterioration	at most 1.10	above 1.10, at most 1.25	above 1.25
Exact-binomial tail coverage	p at least .05 and 10 expected exceptions	p at least .01 but not pass	p below .01
PCA AR spectral radius	below .99	.99 to below 1	at least 1
Invalid decoded paths	zero	above zero, at most 1%	above 1%
Surfaces with IV above 300%	zero	above zero, at most 0.1%	above 0.1%

Diagnostic	Pass (black)	Warn (orange)	Fail (red)
Diagnostic coverage	all requested dates	partial coverage or not run	not a model-quality failure

Only the AR boundary $\rho(A) = 1$ comes from the mathematical stationarity condition. The nearby .99 warning is a practical slow-reversion flag: $\log(1/2)/\log(.99) \simeq 69$ business sessions. Tail cutoffs .05 and .01 are conventional pointwise type-I error levels for the exact binomial coverage test, not Gaussian standard-error approximations. Fewer than ten expected exceptions prevents a black tail result even if the null is not rejected. An IV-ACF comparison with fewer than 30 observed lag pairs likewise cannot receive black.

All remaining boundaries are explicit, round-number engineering tolerances introduced for this retrospective review, not confidence bounds, published universal standards, or thresholds selected on a separate validation set. Frobenius bounds permit 25%/50% norm error; trace bounds permit a factor 1.25/two mismatch; eigen-share bounds permit 10/25 percentage points of summed error; correlation bounds permit .10/.20 absolute error. The spread and forecast tolerances express stated reconstruction goals and allowed deterioration against a simple persistence forecast. They were not used to choose or retrain the models and should be sensitivity-tested before any production acceptance decision.

Next-day forecast in-spread rates and mean outside distances are not graded as reconstruction: a correct conditional expectation need not lie in a narrow realized next-day spread. Price-prediction grading instead compares MAE with the same contracts' origin observed midpoints carried by the origin-only one-day discount factor. Log-IV forecast RMSE is compared with the no-change IV forecast. A ratio at most one means no deterioration; up to 1.10 is the explicit 10% warning band. May/April deterioration compares these persistence-normalized log-IV losses, reducing confusion with changing market volatility. Pinball and joint VaR–ES scores are proper-score comparisons, not zero-centered tests; tail colors refer to coverage, not their raw score magnitudes.

Colors on conditional covariance/trace are descriptive dispersion screens against realized total covariance, not formal conditional-covariance tests; between-origin predicted means can explain a difference. Tail p-values and colors are pointwise and exploratory; no across-model multiplicity correction is claimed. Historical-P comparisons for GROMM-Q are descriptive, not a demand that a Q generator reproduce P outcomes. The complete numeric rules are saved in `TRAFFIC_LIGHT_RULES.json`.

D Surface covariance and spot leverage

The common grid has business-day tenors $(1, 2, 3, 4, 5, 7, 10)$ and NMS $(-1.5, -1, -0.5, 0, 0.5, 1, 1.5)$, giving 49 equally weighted log-IV coordinates. The central common domain avoids comparing different native grids or extreme-wing inversion conventions. All retained reference nodes in this common domain have interior IVs; the broader native reference grid does contain repaired wing nodes. This is not a test of the omitted wings. Native grids remain unchanged.

The fixed three-PC basis uses 252 exact one-session training transitions with endpoints 2023-02-21–2024-03-28, strictly before April. These are eligible retained transitions, not an assertion of 252 consecutive calendar sessions. Their variance shares are 78.19, 10.80, 3.83 percent (total 92.81 percent). No whitening or per-model refitting of this audit basis is used.

Let x_t be the common log-IV vector, U the fixed basis, $\Delta x_{t+1} = x_{t+1} - x_t$, $m_t = E_t[\Delta x_{t+1}]$ and $C_t = \mathrm{Cov}_t(\Delta x_{t+1})$. We compare both $\bar{C}$ and $\bar{C} + \widehat{\mathrm{Cov}}_t(m_t)$ with the sample covariance of observed changes. The latter retains between-origin variation in conditional means; it must not be mistaken for conditional shock dispersion. Across-origin covariance uses denominator $n - 1$; the displayed sum is the fixed-design, sample-covariance convention. Monte Carlo forecast covariance uses the sample convention, whereas exact enumeration of the 252-atom DYSANOS law uses its population denominator 252.

$$E_F = \frac{\|\Sigma_{\text{model}} - \Sigma_{\text{data}}\|_F}{\|\Sigma_{\text{data}}\|_F}.$$

Table 13: Surface covariance errors and variance ratios on each model's available dates. Color rules: Frobenius error: black at most 0.25, orange above 0.25 through 0.50, red above 0.50. Trace ratio: black 0.80–1.25; orange 0.50–2.00 outside that band; red outside 0.50–2.00. These are descriptive review tolerances, not statistical acceptance tests. Missing graded entries are orange; other raw quantities are ungraded.

Model	Days	Frob. cond.	Frob. pooled	Trace cond.	Trace pooled
DYSANOS ordinary (16 factors)	44	0.914	0.821	0.080	0.166
DYSANOS spot-orthogonal (16 factors)	44	0.839	0.775	0.155	0.214
FuNVol	44	0.602	0.580	0.397	0.495
VolGAN PCA8	44	1.000	0.283	0.000251	0.970
AD-Seq-Vol	44	0.665	0.786	0.374	1.542
AD-Seq-Vol-FT	44	0.740	0.697	0.302	1.455
GROMM-P	44	0.398	0.334	0.796	1.268
GROMM-Q	44	0.410	0.280	0.763	1.205
DYSANOS ordinary (16 factors) (tail)	44	0.914	0.820	0.080	0.166
DYSANOS spot-orthogonal (16 factors) (tail)	44	0.839	0.776	0.154	0.214

In Table 13, the two Frobenius columns measure relative covariance-matrix error (zero is exact agreement); the two trace columns divide model by observed total log-IV-change variance (one is scale agreement). Figure 1 plots those same trace columns. Its conditional bar uses the average within-origin forecast covariance; its pooled bar additionally includes between-origin forecast-mean variation. The color bands in the table caption are retrospective engineering review tolerances, not precommitted statistical pass/fail criteria. In particular, the conditional ratio need not equal one because the observed denominator is unconditional.

Table 13 uses each model's available dates. Table 14 restricts every model to the same 44 common origins (2024-04-01–2024-05-31). The PCA basis remains unchanged.

Table 14: Surface covariance comparison restricted to dates available for every model. Color rules: Frobenius error: black at most 0.25, orange above 0.25 through 0.50, red above 0.50. Trace ratio: black 0.80–1.25; orange 0.50–2.00 outside that band; red outside 0.50–2.00. These are descriptive review tolerances, not statistical acceptance tests. Missing graded entries are orange; other raw quantities are ungraded.

Model	Matched days	Frob. pooled	Trace cond.	Trace pooled
DYSANOS ordinary (16 factors)	44	0.821	0.080	0.166
DYSANOS spot-orthogonal (16 factors)	44	0.775	0.155	0.214
FuNVol	44	0.580	0.397	0.495
VolGAN PCA8	44	0.283	0.000251	0.970
AD-Seq-Vol	44	0.786	0.374	1.542
AD-Seq-Vol-FT	44	0.697	0.302	1.455
GROMM-P	44	0.334	0.796	1.268
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DYSANOS ordinary (16 factors) (tail)	44	0.820	0.080	0.166
DYSANOS spot-orthogonal (16 factors) (tail)	44	0.776	0.154	0.214

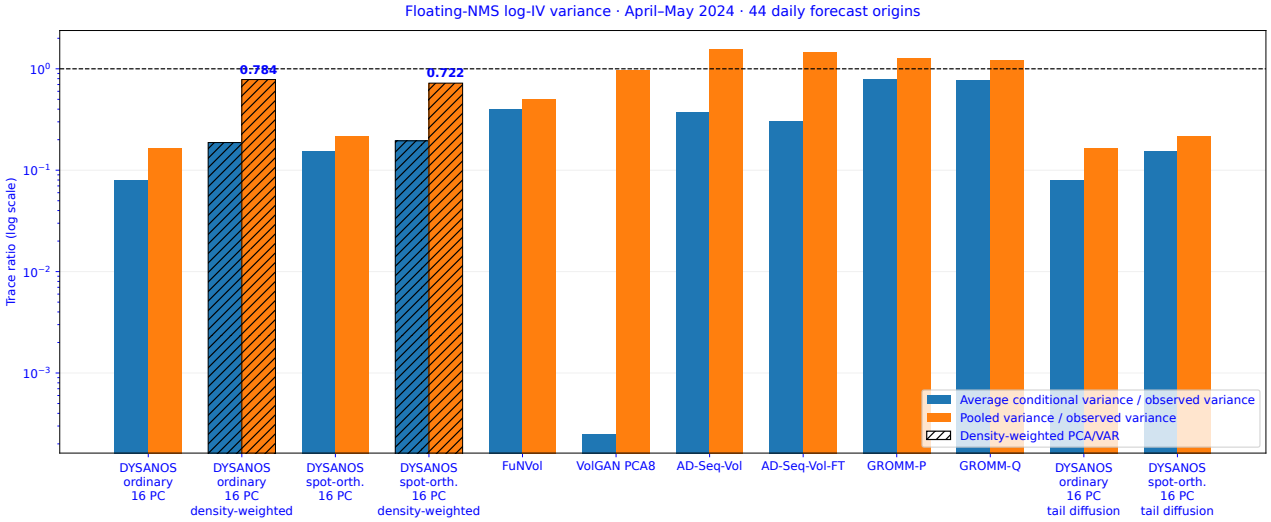


Figure 1: Conditional and pooled surface-variance ratios. The logarithmic scale makes underdispersion visible rather than hiding it behind changing forecast means. Hatched bars add the sixteen-factor density-weighted DYSANOS PCA/VAR experiment on the same 44 origins. Its squared log-DLV errors use causal training-window average grid probability masses; log-forward-vol coordinates keep unit weight. Their remaining diagnostics are reported in the density-weighted follow-up section.

D.1 Why native PCA coverage need not preserve IV variance

The retained DYSANOS PCs minimize Euclidean reconstruction error in 1809 native log-DLV/forward-vol coordinates, not in option prices or log implied volatilities. Locally, if J is the pricing/IV decoder Jacobian, the observable covariance is $J\Sigma_u J'$. A PCA truncation that retains much of $\mathrm{tr}(\Sigma_u)$ need not retain much of $\mathrm{tr}(J\Sigma_u J')$. Drift-adjusting the retained factors for AR fitting can further reduce innovation variance. Floors and nonlinear decoder guards also affect this mapping. Therefore a native coverage percentage is not an observable-variance guarantee.

Table 15: Same-shock variance reconciliation. Each origin uses the same 252 trailing centered native changes and current decoder. Percentages divide by the full centered-native-shock comparator: native energy for the first numeric column, decoded log-IV variance for the others. No shocks are inflated; the comparator is an ablation, not a fitted full-rank alternative. Original decoder guards remain active and their counts are retained. Color rules: Black: ungraded diagnostic quantities. Orange and red grades are not assigned; this ablation is not a calibration pass/fail test. Missing graded entries are orange; other raw quantities are ungraded.

Model	Origin	Native retained %	PCA IV %	AR innovation IV %
DYSANOS ordinary (16 factors)	2024-04-01	82.868	21.410	12.897
DYSANOS ordinary (16 factors)	2024-04-30	83.970	33.061	18.091
DYSANOS ordinary (16 factors)	2024-05-31	84.182	27.966	16.625
DYSANOS spot-orthogonal (16 factors)	2024-04-01	83.026	35.556	28.503
DYSANOS spot-orthogonal (16 factors)	2024-04-30	84.124	46.996	34.264
DYSANOS spot-orthogonal (16 factors)	2024-05-31	84.325	40.326	31.283

D.2 Controlled eight-versus-sixteen-factor DYSANOS comparison

Only the retained rank changes. Both variants use the same 44 origins, 252 exact causal training transitions, empirical paired spot innovations, native states, decoder guards and vector cdxcore IV inversion. Each new rolling basis is aligned with its preceding origin; March 28 supplies a causal alignment seed before scoring starts. The old eight-dimensional subspace is verified to lie inside the

new sixteen-dimensional span at every origin. No shock inflation or tail multiplier is introduced. The separate recursive tests retain the common 32-path, 115-session protocol. This does not resolve the source/reference smoothness difference or change the native density fit.

Table 16: Controlled rank comparison on all 44 origins. Native coverage is averaged across origins; decoded covariance is compared against the unchanged observable reference. Counts, native coverage and covariance scale are distinct quantities. Color rules: Frobenius error: black at most 0.25, orange above 0.25 through 0.50, red above 0.50. Trace ratio: black 0.80–1.25; orange 0.50–2.00 outside that band; red outside 0.50–2.00. These are descriptive review tolerances, not statistical acceptance tests. AR spectral radius: black below 0.99, orange from 0.99 to below 1, red at least 1. Native coverage and counts are ungraded. Missing graded entries are orange; other raw quantities are ungraded.

Variant / rank	Days	Native %	Trace cond.	Trace pooled	Frob. pooled	Max radius
Ordinary / 8	44	70.598	0.037	0.059	0.939	0.699
Spot-orthogonal / 8	44	70.770	0.132	0.152	0.843	0.695
Ordinary / 16	44	83.773	0.080	0.166	0.821	0.735
Spot-orthogonal / 16	44	83.862	0.155	0.214	0.775	0.730

Table 17: Eight- and sixteen-factor same-shock ablations. Percentages use the same full native-shock comparator at each origin. Decoder guards remain active, so this is not an attribution to linear PCA alone. Color rules: Black: ungraded same-shock ablation quantities. Orange and red grades are not assigned; variance retention is not a calibration pass/fail test. Missing graded entries are orange; other raw quantities are ungraded.

Variant / rank	Origin	Native retained %	PCA IV %	AR innovation IV %
Ordinary / 8	2024-04-01	68.928	7.536	4.880
Ordinary / 8	2024-04-30	71.036	14.034	8.493
Ordinary / 8	2024-05-31	71.392	14.754	9.047
Ordinary / 16	2024-04-01	82.868	21.410	12.897
Ordinary / 16	2024-04-30	83.970	33.061	18.091
Ordinary / 16	2024-05-31	84.182	27.966	16.625
Spot-orthogonal / 8	2024-04-01	69.203	26.962	23.991
Spot-orthogonal / 8	2024-04-30	71.334	36.093	29.755
Spot-orthogonal / 8	2024-05-31	71.667	32.473	26.499
Spot-orthogonal / 16	2024-04-01	83.026	35.556	28.503
Spot-orthogonal / 16	2024-04-30	84.124	46.996	34.264
Spot-orthogonal / 16	2024-05-31	84.325	40.326	31.283

Full rank-comparison results, including PC1 tail counts, recursive invalid/extreme rates and decoder clipping counts, are retained in `rank_comparison.json`. Native energy retention and a stable AR matrix do not by themselves establish observable covariance calibration.

Leading eigenvalue shares use the full covariance trace as denominator. Principal angles compare the leading three-dimensional subspaces of the full observable covariance; the table gives the maximum angle in degrees. The full machine-readable summary also contains the three-PC projected covariance comparison. Using three dimensions for a three-dimensional projected subspace angle would be vacuous, so only the full-space angles are interpreted.

Table 18: Log-IV forecast errors and leading covariance-subspace agreement. Color rules: RMSE: black at most the persistence RMSE, orange above persistence through 1.10 times persistence, red above 1.10 times. Maximum angle: black at most 15 degrees, orange above 15 through 30, red above 30. Eigen-share L1 error: black at most 0.10, orange above 0.10 through 0.25, red above 0.25. Missing graded entries are orange; other raw quantities are ungraded.

Model	Forecast RMSE	Persistence RMSE	Max angle	Share L1 error
DYSANOS ordinary (16 factors)	0.174	0.190	79.259	0.151
DYSANOS spot-orthogonal (16 factors)	0.176	0.190	61.570	0.127
FuNVol	0.183	0.190	48.536	0.134
VolGAN PCA8	0.215	0.190	48.630	0.130
AD-Seq-Vol	0.237	0.190	51.960	0.096
AD-Seq-Vol-FT	0.266	0.190	52.164	0.077
GROMM-P	0.120	0.190	85.623	0.102
GROMM-Q	0.125	0.190	85.754	0.111
DYSANOS ordinary (16 factors) (tail)	0.174	0.190	72.909	0.154
DYSANOS spot-orthogonal (16 factors) (tail)	0.176	0.190	58.317	0.128

RMSE is in log-IV units over scored dates and all 49 cells. Persistence predicts no surface change. RMSE values concern raw model forecasts, not the artificial demeaning used by some hedge optimizers.

Spot leverage is $\mathrm{Corr}(r_{t+1}, U_j^\top \Delta x_{t+1})$, where r is the one-session forward-normalized log return. Signs are fixed by the training basis. Model values below use pooled joint moments; per-component absolute errors are in the JSON. A native DLV loading is not substituted for this observable correlation.

Table 19: Pooled correlations between spot returns and fixed-basis observable PC changes. Color rules: Absolute correlation error versus the observed reference: black at most 0.10, orange above 0.10 through 0.20, red above 0.20. Reference correlations and counts are neutral. Missing graded entries are orange; other raw quantities are ungraded.

Model	Days	Spot/PC1	Spot/PC2	Spot/PC3
Observed reference	44	-0.680	0.157	-0.008
DYSANOS ordinary (16 factors)	44	-0.149	0.035	0.029
DYSANOS spot-orthogonal (16 factors)	44	-0.694	0.729	0.568
FuNVol	44	-0.711	0.457	0.232
VolGAN PCA8	44	-0.451	0.338	0.534
AD-Seq-Vol	44	-0.269	0.318	0.263
AD-Seq-Vol-FT	44	-0.340	0.436	0.311
GROMM-P	44	-0.250	0.157	0.061
GROMM-Q	44	-0.214	0.149	0.076
DYSANOS ordinary (16 factors) (tail)	44	-0.147	0.046	0.052
DYSANOS spot-orthogonal (16 factors) (tail)	44	-0.693	0.740	0.566

The JSON additionally separates correlations from the average conditional joint covariance and correlations of realized forecast residuals (observed changes minus each model's predicted means). This prevents between-origin mean variation from being mistaken for conditional spot leverage.

E Bid–ask fit and next-day prediction

For quoted bid b, ask a and predicted price p, outside distance is $d = \max(b - p, p - a, 0)/(a - b)$. We report the in-spread share, mean and 95th percentile of d, and cash-price MAE. Crossed/zero-width/nonfinite inputs are excluded with counts. Quotes are licensed observed bid/ask, not fabricated spreads around fitted prices.

Table 20: Same-day quoted-spread reconstruction and next-day conditional-mean price prediction. Color rules: Same-day in-spread share: black at least 95%, orange 80% to below 95%, red below 80%. Forecast MAE / same-contract persistence MAE: black at most 1.00, orange above 1.00 through 1.10, red above 1.10. Other forecast statistics are ungraded; next-day in-spread share is not a reconstruction test. Missing graded entries are orange; other raw quantities are ungraded.

Model	Recon. in	Pred. N	Pred. in	Mean distance	MAE / spot	MAE ratio
DYSANOS ordinary (16 factors)	0.744	44	0.013	36.830	0.002	1.009
DYSANOS spot-orthogonal (16 factors)	0.744	44	0.010	37.309	0.002	1.011
FuNVol	–	44	0.013	43.205	0.002	1.047
VolGAN PCA8	–	44	0.006	139.538	0.005	2.785
AD-Seq-Vol	–	44	0.009	72.848	0.002	1.331
AD-Seq-Vol-FT	–	44	0.007	69.679	0.002	1.284
GROMM-P	0.928	44	0.009	43.425	0.002	1.009
GROMM-Q	0.928	44	0.009	41.915	0.002	0.995
DYSANOS ordinary (16 factors) (tail)	0.744	44	0.012	36.714	0.002	1.010
DYSANOS spot-orthogonal (16 factors) (tail)	0.744	44	0.011	37.114	0.002	1.011

Reconstruction uses all available quoted sides with origin tenors 1–10D and $|\text{NMS}| \leq 1.5$ for DYSANOS and GROMM. Conditional IV generators do not have an independently fitted same-day observable decoder in this audit: their reconstruction is n/a, not perfect by copying their conditioning input. GROMM's stored 252-transition fit percentages are not substituted for today's measured quoted reconstruction.

Next-day prediction uses surviving OTM hedge-book contracts with origin tenors 2–5D and $|\text{NMS}| \leq 1.5$, matched at their exact cash strike, expiry and settlement type. The expected raw scenario price is compared with tomorrow's observed spread. Financing/carry is fixed from the origin; tomorrow's spot/forward curve is not inserted into the forecast. Expiring 1D contracts are settlements, not next-day quotes. This is a limited retained-book forecast population, not every model-grid contract. Population, matched and excluded counts and moneyness-bucket scores are retained per date. Reconstruction and forecast columns have different populations and should not be compared as like-for-like fit rates.

A correct stochastic model need not put its conditional-mean forecast inside a narrow realized next-day spread. The forecast in-spread rate is a prediction diagnostic, not an arbitrage criterion. Quote aggregates give equal weight to dates, not to the number of contracts on each date. All daily counts, mean and 95th-percentile outside distances, price MAEs and buckets are retained in the CSV/JSON outputs.

`iv_rmse_buckets.csv` separately reports both log-IV RMSE and annual-IV RMSE for the explicitly defined point forecast $\hat{\sigma}_{t+1} = \exp(x_t + m_t)$, anchored to the common observed origin surface. This is a geometric-IV point forecast, not $E_t[\sigma_{t+1}]$ and not IV obtained by inverting a mean option price. Errors are bucketed by expiry and NMS. No realized forward is used to manufacture an IV from a potentially non-arbitrage-bounded mean price.

F Out-of-sample window stability

Every available saved origin is scored. Monthly RMS errors are separated by April and May; no six-month evaluation is claimed. Both monthly and 21-origin rolling RMSE take the square root after averaging squared daily RMSE, consistent with the full-window metric. Rolling deterioration subtracts the first 21-origin value. A 63-session deterioration estimate is unavailable. These overlapping summaries are descriptive, not independent replications. Forecast horizon (one session), surface maturity (1–10 sessions) and evaluation-window length (44 origins) remain separate.

Table 21: Monthly next-day log-IV forecast RMSE and evaluated origins. Color rules: Monthly RMSE / monthly persistence RMSE: black at most 1.00, orange above 1.00 through 1.10, red above 1.10. May/April persistence-normalized deterioration: black at most 1.10, orange above 1.10 through 1.25, red above 1.25. Counts are neutral. Missing graded entries are orange; other raw quantities are ungraded.

Model	April days	April RMSE	May days	May RMSE	Skill deterioration
DYSANOS ordinary (16 factors)	22	0.161	22	0.187	1.127
DYSANOS spot-orthogonal (16 factors)	22	0.164	22	0.187	1.105
FuNVol	22	0.183	22	0.184	0.973
VolGAN PCA8	22	0.234	22	0.193	0.800
AD-Seq-Vol	22	0.256	22	0.216	0.819
AD-Seq-Vol-FT	22	0.311	22	0.211	0.659
GROMM-P	22	0.128	22	0.110	0.835
GROMM-Q	22	0.140	22	0.107	0.744
DYSANOS ordinary (16 factors) (tail)	22	0.161	22	0.187	1.125
DYSANOS spot-orthogonal (16 factors) (tail)	22	0.164	22	0.186	1.102

G Tail risk calibration

Targets are the three observable PC changes and spot log return, not hedge P&L. Quantile levels are 1, 5, 95 and 99 percent. For q_α, pinball loss is $(\alpha - \mathbf{1}_{\{y<q_\alpha\}})(y - q_\alpha)$. Lower-tail exceptions are $y < q_\alpha$ and upper-tail exceptions $y > q_\alpha$. The outputs contain exception rates, exact binomial coverage intervals and p-values, asymptotic Kupiec p-values and, when identifiable, Christoffersen independence p-values. Clustering tests are n/a when exception histories do not identify both transition rows.

The following counts are for PC1; all four targets and all probabilities are in `tail_scores.csv`. Expected exception counts with 44 dates are 0.44 in each 1% tail and 2.2 in each 5% tail. Every tail test is therefore labelled exploratory. Ten thousand Monte Carlo scenarios do not create ten thousand held-out market dates. No green pass is awarded from a nonsignificant p-value. GROMM-Q is compared with historical outcomes but is not thereby tested for physical-law calibration.

Table 22: Exploratory PC1 tail-forecast exception counts and lower-5-percent pinball loss. Color rules: Exception-count colors use pointwise two-sided exact-binomial coverage p-values: red below 0.01; orange from 0.01 to below 0.05, or whenever expected exceptions are fewer than 10; black requires p at least 0.05 and at least 10 expected exceptions. Pinball loss is ungraded. Missing graded entries are orange; other raw quantities are ungraded.

Model	Days	Below 1%	Below 5%	Above 95%	Above 99%	5% pinball
DYSANOS ordinary (16 factors)	44	8	13	10	9	0.218
DYSANOS spot-orthogonal (16 factors)	44	7	11	8	7	0.196
FuNVol	44	2	4	9	3	0.115
VolGAN PCA8	44	34	34	10	10	0.810
AD-Seq-Vol	44	12	15	11	5	0.258
AD-Seq-Vol-FT	44	5	8	21	10	0.156
GROMM-P	44	0	0	1	0	0.096
GROMM-Q	44	0	0	2	1	0.106
DYSANOS ordinary (16 factors) (tail)	44	10	14	9	7	0.233
DYSANOS spot-orthogonal (16 factors) (tail)	44	4	10	9	7	0.177

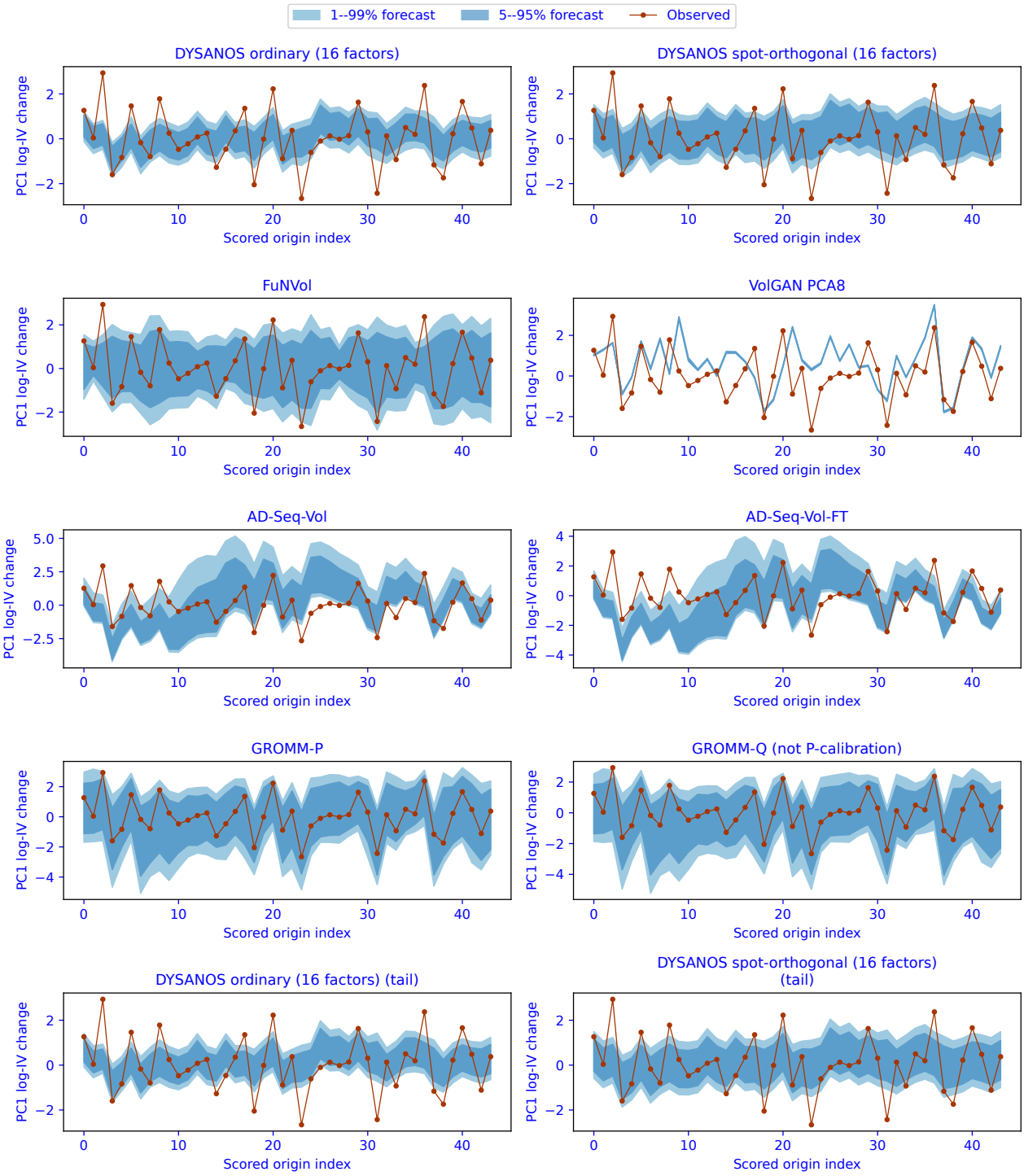


Figure 2: Raw PC1 forecast bands and observed changes. Narrow forecast bands can reveal severe conditional underdispersion even when pooled covariance looks realistic.

The lower 5% ES estimate is the Monte Carlo mean below the empirical 5% quantile. The JSON contains the joint Fissler–Ziegel score, with $G_1 = 0$, $G_2(e) = \exp(e)$, for signed lower-tail ES:

$$S_\alpha(y, q, e) = e^e \left[e - q + \frac{\mathbf{1}_{\{y \le q\}}}{\alpha}(q - y) - 1 \right].$$

Lower expected score is better for a fixed target and units. Unlike the negative-ES-only FZ0 specialization, this choice does not discard forecasts whose lower-tail ES is positive. Scores of differently scaled targets are not pooled into a headline ranking.

H PCA–AR(1) stationarity and floating-IV persistence

The analytic stationarity question is restricted to the fitted floating-volatility PCA backbone, with coefficients frozen at each origin. Write its transition as

$$h_{n+1} = b + Ah_n + \varepsilon_{n+1}, \qquad E[\varepsilon_{n+1}] = 0, \quad \mathrm{Cov}(\varepsilon_{n+1}) = Q. \tag{1}$$

Innovations are independent across sessions under the fitted law and have finite covariance. A Gaussian stock-return innovation may contribute to ε; it changes Q, including any stock–factor cross-covariance, not the stability criterion. No stationarity condition is imposed on the stock price. A sufficient condition for a unique causal stationary backbone, and the criterion tested here, is

$$\rho(A) < 1, \qquad \bar{h} = (I - A)^{-1}b, \qquad V = AVA^{\top} + Q. \tag{2}$$

The Lyapunov solution V is the stationary covariance. If the complete innovation vector is jointly Gaussian, the stationary law is $N(\bar{h}, V)$; Gaussianity is not needed for covariance stationarity. For the orthogonal variant, the innovation covariance includes the stock loading projected onto the 16 PCA coordinates. Starting a forecast at the observed state need not start it in its invariant distribution: the stable backbone converges toward that law. This is a per-fit assessment, not a claim that a daily re-estimated sequence has constant parameters.

Let $z_{j,t} = U_j^{\top} x_t$. The PCA basis is fixed before April, while each model may continue using its own daily rolling factors internally. We never concatenate native factor scores across changing bases. We separately retain level ACF, change ACF and expiry-by-expiry ATM log-IV level ACF. Exact session separations are used; lag 21 has 23 observed pairs and lag 63 has none.

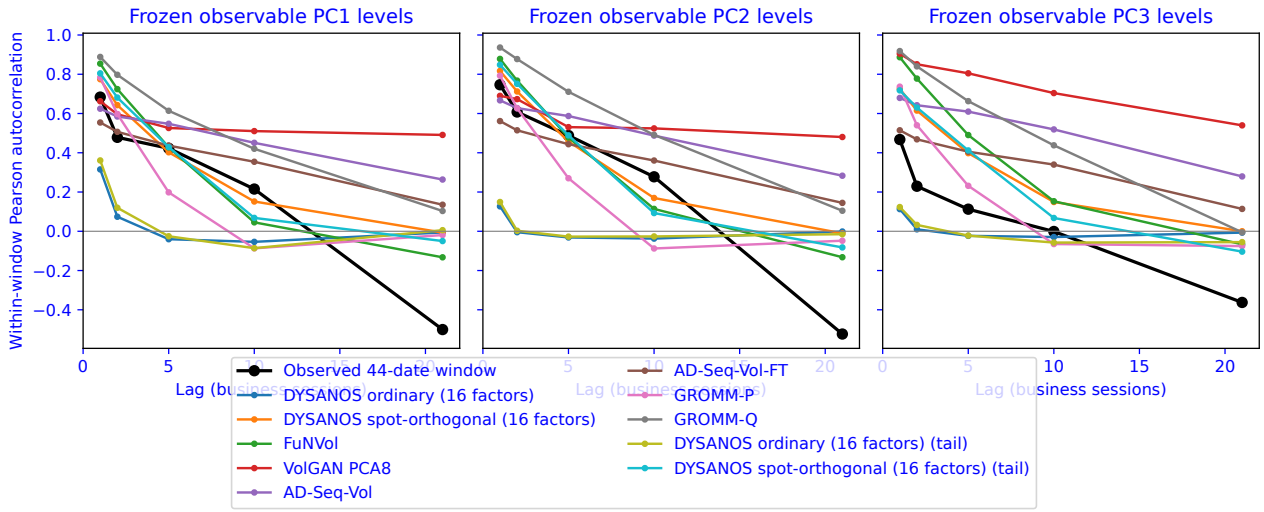


Figure 3: PCA-level autocorrelation decay. Model curves average within-path Pearson ACFs from the first 44 simulated sessions, matching the observational window length; they are not formed by concatenating different dates' rolling models. The full 128-session checks are stored separately. Short-window detrending can itself produce negative long-lag estimates.

Table 23: Maximum absolute floating-IV ACF discrepancy across the three fixed observable PCs. Lag 21 has only 23 observed pairs, so cannot receive black. Color rules: Maximum absolute PC-ACF error: black at most 0.10, orange above 0.10 through 0.20, red above 0.20. Fewer than 30 observed lag pairs forces at least orange, even for a small error. Missing graded entries are orange; other raw quantities are ungraded.

Model	Lag 1 error	Lag 5 error	Lag 21 error
DYSANOS ordinary (16 factors)	0.620	0.519	0.522
DYSANOS spot-orthogonal (16 factors)	0.258	0.285	0.513
FuNVol	0.419	0.378	0.391
VolGAN PCA8	0.437	0.692	1.004
AD-Seq-Vol	0.212	0.497	0.807
AD-Seq-Vol-FT	0.185	0.293	0.668
GROMM-P	0.269	0.224	0.481
GROMM-Q	0.451	0.550	0.628
DYSANOS ordinary (16 factors) (tail)	0.597	0.515	0.509
DYSANOS spot-orthogonal (16 factors) (tail)	0.250	0.299	0.451

For DYSANOS only, we report the saved backbone spectral radius, modal damping half-lives $\log(1/2)/\log|\lambda_j|$ for stable nonzero eigenvalues, and innovation variance traces at horizons 1,5,21,63. Complex/negative eigenvalues describe oscillatory damping, not a monotone scalar half-life. Half-lives are measured in business sessions. Stationary means, covariances and numerical Lyapunov residuals are retained in the dated JSON files. All 88 fitted radii are already below one before the configured spectral-radius cap is applied.

Table 24: Stationarity of the floating-volatility PCA–AR(1) backbones; half-lives are in business sessions. Color rules: AR spectral radius: black below 0.99, orange from 0.99 to below 1, red at least 1. Stable-fit counts and half-lives are ungraded. Missing graded entries are orange; other raw quantities are ungraded.

DYSANOS backbone	Stable fits	Min radius	Max radius	Max half-life
DYSANOS ordinary (16 factors)	44/44	0.565	0.735	2.246
DYSANOS spot-orthogonal (16 factors)	44/44	0.574	0.730	2.201

The separate recursive tests below concern decoded floating IVs, not stationarity of spot and not an additional acceptance condition for the PCA–AR backbone. They retain the original model laws without substituting Gaussian stock shocks or changing fitted coefficients.

Table 25: Recursive floating-IV diagnostics: evaluated origins and extreme/invalid rates. Color rules: Extreme-surface rate (any common-grid IV above 300%): black zero, orange above zero through 0.1%, red above 0.1%. Invalid-path rate: black zero, orange above zero through 1%, red above 1%. Evaluated-origin counts are neutral. Missing graded entries are orange; other raw quantities are ungraded.

Model	Recursive origins	Extreme surface rate	Invalid path rate
DYSANOS ordinary (16 factors)	44	0.000	0.000
DYSANOS spot-orthogonal (16 factors)	44	0.000	0.001
FuNVol	44	0.000	0.000
VolGAN PCA8	44	0.767	0.000
AD-Seq-Vol	44	0.000	0.000
AD-Seq-Vol-FT	44	0.000	0.000
GROMM-P	44	0.000	0.000
GROMM-Q	44	0.000	0.000
DYSANOS ordinary (16 factors) (tail)	44	0.000	0.000
DYSANOS spot-orthogonal (16 factors) (tail)	44	0.000	0.000

An extreme surface has at least one common-grid annual IV above 3.0 (300%). All option-price inversion uses vectorized `cdxcore.bs.implied` on raw, unclipped inputs, with its 10^{-8} price tolerance and an independent vectorized `bs.price` repricing check. The former additional time-value/interiority veto has been removed. The library's default 1%–500% volatility bounds and intrinsic-price conventions are retained and counted explicitly; a library boundary estimate is not called a failed model path solely for having tiny time value. Native-price bound violations and repricing residuals remain recorded. These numerical point estimates do not imply uniquely identified wing volatilities at sub-tolerance prices. A completed test with no usable complete IV paths remains in coverage and invalid-rate statistics but has no ACF. All DLV clipping counts are retained separately. Recursive paths use frozen fits and their original innovation laws; finite-horizon survival is not a non-explosion proof.

I Density-weighted DYSANOS: completed follow-up

All 44 April–May origins have one-day surface/tail/leverage forecasts, quoted-contract forecasts and frozen-fit recursive tests for both weighted variants. Each recursive test uses 32 paths, 115 business sessions and seed 20261111, exactly as for the baselines. Existing baseline fits and diagnostics are retained, not replaced. The separate density-prefixed CSVs contain all target, tenor, moneyness, monthly and rolling-window scores.

For each origin we average decoded grid probability masses over the 252 training origins, writing $w_{jk} = \overline{p}^j_k$. These are node probabilities, not probability-density heights on the nonuniform strike grid. Put $D = \mathrm{diag}(\sqrt{w})$ for log-DLV-squared coordinates, and unit diagonal entries for log-forward-volatility coordinates. Both PCA and VAR are fitted to $z = Du$; simulated changes are mapped back with the same frozen D. No probability floor or shock multiplier is added. A zero-weight native coordinate is frozen. Rolling orientations map the preceding dated loadings into the new causal metric.

In the spot-orthogonal variant, $\beta = \mathrm{Cov}(\Delta z, r)/\mathrm{Var}(r)$ is removed before PCA and restored in simulation. With $h = U'(z - \bar{z})$, the update is

$$u_{t+1} = u_t + D^+\{U[b + (A - I)h_t + \varepsilon_{t+1}] + \beta r_{t+1}\}.$$

The paired empirical (r, ε) law preserves all spot–factor cross-covariances, including residual dependence after VAR fitting. Ordinary PCA does not project out spot but keeps that joint dependence. Fixed-cash-strike option repricing separately uses the updated spot and forward. Across all 44 orthogonal fits, the pre-PCA spot-orthogonality error is below $4 \cdot 10^{-17}$; paired replay and covariance identities agree to about 10^{-15}. The reported predictive variance already includes the stock contribution.

Table 26: Weighted and unweighted one-day observable covariance comparison, plus GROMM references, on the same 44 origins. RMSE here is ungraded; native and weighted PCA coverage have different denominators. Color rules: Frobenius error: black at most 0.25, orange above 0.25 through 0.50, red above 0.50. Trace ratio: black 0.80–1.25; orange 0.50–2.00 outside that band; red outside 0.50–2.00. These are descriptive review tolerances, not statistical acceptance tests. Missing graded entries are orange; other raw quantities are ungraded.

Model	Trace cond.	Trace pooled	Frob. pooled	RMSE
Ordinary / unweighted	0.080	0.166	0.821	0.174
Ordinary / density	0.187	0.784	0.238	0.124
Spot-orthogonal / unweighted	0.155	0.214	0.775	0.176
Spot-orthogonal / density	0.195	0.722	0.310	0.120
GROMM-P	0.796	1.268	0.334	0.120
GROMM-Q	0.763	1.205	0.280	0.125

A larger conditional bar in Figure 1 is not automatically better calibrated. To remove the

unconditional-variance denominator, also compare mean forecast variance with realized squared forecast errors:

$$\mathcal{V}_{\text{err}} = \frac{\operatorname{tr}(\overline{C})}{n^{-1}\sum_t \|\Delta x_{t+1} - m_t\|^2}.$$

One is the population reference if the conditional mean and covariance are correct. This is a descriptive 44-origin comparison, not an independent-sample acceptance test. Misspecified means can also distort it. Weighted DYSANOS is still narrow on this measure; GROMM is broader than the observed forecast errors. Hence neither a larger conditional trace nor a pooled ratio near one alone establishes superior uncertainty calibration.

Table 27: Predictive variance versus squared forecast-error scale. All values are ungraded descriptive diagnostics; no orange/red thresholds are assigned. Color rules: Black: ungraded entries. No orange or red grades are assigned. Missing graded entries are orange; other raw quantities are ungraded.

Model	Predicted trace	Forecast MSE	Variance / MSE
Ordinary / unweighted	0.145	1.487	0.098
Ordinary / density	0.340	0.756	0.450
Spot-orthogonal / unweighted	0.281	1.513	0.186
Spot-orthogonal / density	0.354	0.700	0.506
GROMM-P	1.446	0.701	2.064
GROMM-Q	1.386	0.763	1.816

Table 28: Observed-quote tests. Same-day reconstruction is unchanged because the daily decoder was not refitted. Next-day scores use identical surviving 2–5D cash contracts; normalized price MAEs and their reference are averaged equally over origins. Color rules: Same-day in-spread share: black at least 95%, orange 80% to below 95%, red below 80%. Forecast MAE / same-contract persistence MAE: black at most 1.00, orange above 1.00 through 1.10, red above 1.10. Other forecast statistics are ungraded; next-day in-spread share is not a reconstruction test. Missing graded entries are orange; other raw quantities are ungraded.

Model	Recon. inside	Quote count	Forecast MAE / spot	MAE / persistence
Ordinary / density	0.744	7610	0.002	1.018
Spot-orthogonal / density	0.744	7610	0.002	1.012

Table 29: Observable PC1 tail exceptions and lower-tail pinball score. All PC1–PC3 and spot quantiles, coverage/independence tests and joint VaR–ES scores are retained in the density-weighted JSON and CSV files. Expected counts are only 0.44 or 2.2 per tail. Color rules: Exception-count colors use pointwise two-sided exact-binomial coverage p-values: red below 0.01; orange from 0.01 to below 0.05, or whenever expected exceptions are fewer than 10; black requires p at least 0.05 and at least 10 expected exceptions. Pinball loss is ungraded. Missing graded entries are orange; other raw quantities are ungraded.

Model	1%	5%	95%	99%	5% pinball
Ordinary / density	3	10	4	3	0.099
Spot-orthogonal / density	3	6	5	2	0.090

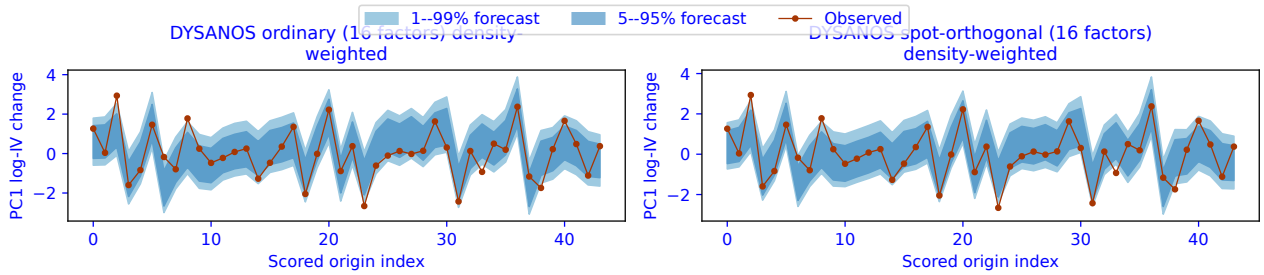


Figure 4: Weighted-model PC1 forecast bands and realized changes, using the same pre-April observable PCA basis.

April/May summaries and rolling 21-session deterioration are reported in the density-prefixed monthly/rolling CSVs. The 44-origin window does not identify 63-session observed deterioration or ACF, and 1% tail tests remain exploratory.

Table 30: Weighted floating-volatility PCA–VAR backbone stationarity. Half-lives are in business sessions; no stationarity claim is made for spot levels or for the whole daily re-estimated model sequence. Color rules: AR spectral radius: black below 0.99, orange from 0.99 to below 1, red at least 1. Stable-fit counts and half-lives are ungraded. Missing graded entries are orange; other raw quantities are ungraded.

Model	Stable fits	Min radius	Max radius	Max half-life
Ordinary / density	44/44	0.902	0.925	8.929
Spot-orthogonal / density	44/44	0.903	0.926	9.042

Table 31: Maximum absolute ACF errors across the three fixed observable PC-level scores. Modeled ACF uses each path's first 44 sessions, not concatenated rolling native factors. Color rules: Maximum absolute PC-ACF error: black at most 0.10, orange above 0.10 through 0.20, red above 0.20. Fewer than 30 observed lag pairs forces at least orange, even for a small error. Missing graded entries are orange; other raw quantities are ungraded.

Model	Lag 1	Lag 5	Lag 21
Ordinary / density	0.097	0.270	0.449
Spot-orthogonal / density	0.106	0.251	0.448

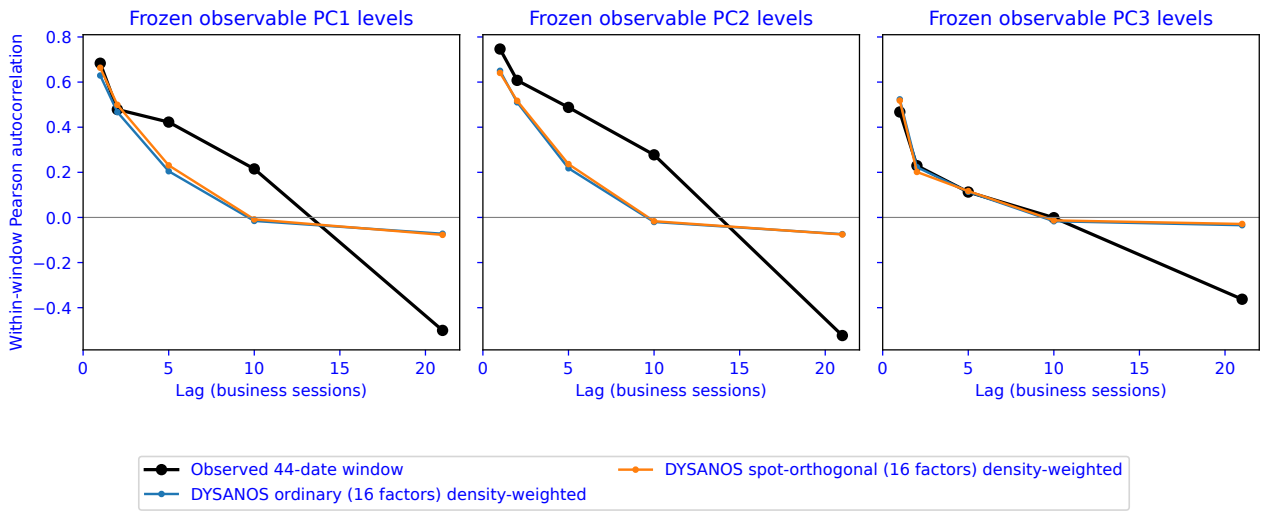


Figure 5: Weighted-model observable-PC persistence against the identical 44-origin observed window. Frozen native metrics are inverted before extracting common-grid IVs.

Table 32: Complete 32-path, 115-session recursive tests on all 44 origins. Rates are origin averages. Clips count individual native coordinates and remain disclosed even when decoded IVs are valid; finite-horizon survival is not a non-explosion proof. Color rules: Extreme-surface rate (any common-grid IV above 300%): black zero, orange above zero through 0.1%, red above 0.1%. Invalid-path rate: black zero, orange above zero through 1%, red above 1%. Evaluated-origin counts are neutral. Missing graded entries are orange; other raw quantities are ungraded.

Model	Origins	Invalid paths	Extreme surfaces	Lower clips	Upper clips
Ordinary / density	44	0.000	0.000	14273189	615885
Spot-orthogonal / density	44	0.000	0.000	16827255	1334816

Implementation feedback: density weighting largely restores forward-volatility variation lost by Euclidean native PCA, but numerical DLV floors/guards and the 0.1-input versus 0.25-observable-reference smoothing difference remain. This follow-up changes the fitted reduction metric, not the pricing geometry, spot law or hedge optimizer. Large fitted metrics and recursive arrays remain external; aggregates and checksummed loading manifests are retained.

J Model-specific implementation choices and difficulties

This appendix is AI-agent implementation feedback based on actual retained code and artifacts, not a claim that these choices are prescribed by the cited papers. Existing model laws and fits are unchanged; the additional May FT fits use the same rolling recipe and are separately identified. The audit does not treat legacy hedging results as newly validated.

J.1 Our DYSANOS fits: ordinary and spot-orthogonal

Both variants use the retained nine tenors $(1, 2, 3, 4, 5, 10, 15, 20, 25)$, 200 interior strikes and native NMS $(-4, 3)$. Daily source prices came from saved 0.1-SANOS surfaces; the representation itself has zero backbone smoothing. Native coordinates contain nine blocks of 200 log-DLV-squared coordinates plus a log-forward-volatility coordinate (1809 in total). This is not the later requested 13-tenor correction, and not the former 1811-coordinate smoothed SANOS state.

Each daily fit uses 252 eligible exact-session transitions, rank 16, ridge 10^{-6} and spectral-radius

cap 0.999. Factor orientation is aligned to the preceding fit. The raw and constrained radii are both saved. The orthogonal variant first removes the spot projection from state changes and adds its loading back during simulation. The analytic stationarity score tests the 16 floating-volatility AR coordinates only; its innovation covariance includes the projected stock contribution and cross-covariances, without testing stock-price stationarity.

The legacy conversion contains zero-over-zero/unidentified and floored DLV coordinates. The report preserves their counts rather than explaining them away as successful calibration. The model decoder retains local-volatility bounds 0.001–1000 and the numerical log-strike bound 30; it records clipping. For one-day forecasts we enumerate all 252 paired empirical innovations exactly. These repeated historical shocks are not 10,000 independent observations. Spot follows the original residual bootstrap here, not a newly imposed front-density marginal. These defects and choices limit the interpretation of this legacy model.

Across the 44 origins, the mean native 16-PC variance coverage is 83.77 percent for ordinary PCA and 83.86 percent after spot orthogonalisation. The daily conversion has 296–453 floored nodes out of 1800 DLV coordinates (median 390). Native-coordinate coverage is not the same denominator as observable log-IV-change coverage.

J.2 Additional causal Tail Diffusion cases

Both ordinary and spot-orthogonal variants retain the same surface decoder and 16-factor AR coefficients but replace empirical surface-innovation replay by the paper's scalar log-variance OU scale mixture. Normal spot innovations are separated by the fitted joint covariance; tail scaling applies to the spot-conditioned Gaussian surface residual. Each dated tail law uses only that fit's preceding 252 exact-session innovations, resetting the latent filter across gaps. Its likelihood uses the existing Gaussian approximation to log-chi-square radius measurements, here in 16 dimensions rather than the paper's 200-dimensional application. This approximation is an explicit limitation.

With fitted conditional factor loading L, the surface shock is $\sqrt{\rho \exp(g - v_g/2)}Lz$, while $g^+ = \phi g + \sqrt{v_g(1-\phi^2)}\xi$. The existing tail fitter estimates (ρ, ϕ, v_g), adding the previous origin's parameters as an optimizer start. The initial latent law is the causal one-step predictive distribution from the trailing radius filter, not an unconditional stationary reset at every simulated session. No April–May future observation tunes these parameters. One-day forecasts use 10,000 draws and recursive tests the same 32 paths over 115 sessions. Tail thickness is not artificially rescaled to match test-window variance.

At this build there are 88 tail fits and 1 raw optimizer warnings. 1 warning cases have independent projected-gradient verification at 10^{-6} mean-log-likelihood accuracy, without changing parameters. Exact original messages remain in the fitted records and review sidecars; a local gradient check is not a global-optimum certificate.

J.3 FuNVol

This is the single-asset adaptation on the seven-tenor, 19-NMS-node short-end grid. Inverse-softplus IVs are smoothed on a fourth-order Legendre basis, then reduced to three functional PCs. Joint score/log-spot dynamics use separate drift and diffusion GRUs: three layers, width 16, ten-state history, Gaussian increments over 1/252 year. The FPC basis and normalization are inherited through the rolling checkpoint chain, not recomputed daily. A 2022 comparison selected 100 rolling epochs; subsequent daily fits used all their causal window rather than a future holdout.

The first causal April fit generated spot volatility 0.333% against trailing historical 0.778%, failing the original 0.5-ratio dispersion gate. The user explicitly authorized proceeding with those numbers. The warning remains part of the model interpretation; no calibration pass is retroactively claimed. Softplus ensures positive IV, not absence of static/dynamic arbitrage. The earlier narrow static-arbitrage screen covered only 11 strikes within approximately 0.5% of ATM. The present central-grid

audit is different and is not an arbitrage proof.

The retained first-April FPC basis reports 97.849% cumulative training variance coverage for its three components. This is variance of Legendre-smoothed inverse-softplus IV levels in its initializer coordinate system, not variance coverage of daily observable log-IV changes in the audit. These percentages must not be compared without their denominators.

Reference: Choudhary, Jaimungal and Bergeron, FuNVol, Quantitative Finance 24 (2024), DOI 10.1080/14697688.2024.2396977.

J.4 VolGAN PCA8

The retained model is the paper-inspired eight-PC adaptation, not the original full-surface VolGAN. Separate train-only bases compress log-IV levels in the condition and log-IV changes in the output. Spot return remains an explicit output. Frozen initializer bases are inherited across dates; network weights and latent normalization are updated causally. Conditioning includes two previous annualized returns and trailing 21-session realized volatility. Softplus networks use noise dimension 32 and hidden dimension 16.

The 2022 initializer trained for 10,000 epochs; selected rolling updates use 300 epochs, and 10,000 sampled scenarios per origin is a different quantity. PCA bases and scaling use only training data. The gradient-matching phase continues from updated weights rather than restarting; its 25 epochs are included in the training total. The initializer selected rolling-update settings on a shorter 200-observation 2022 window because fewer than 252 observations were available. The native-model and short-end-data adaptations must not be described as an exact reproduction of published results.

The one-day surface audit re-samples the saved, version-checked checkpoints on CPU with seed 20261111, preserving the model's decoding and conditioning. Recursive sampling updates each path's own IV surface, recent returns and realized-volatility condition; it does not feed actual future market data back in. Conditional collapse and recursive extreme IVs are reported, not repaired by rescaling shocks.

Reference: Vuletic and Cont, VolGAN, DOI 10.1080/1350486X.2025.2471317.

J.5 AD-Seq-Vol and fine-tuned AD-Seq-Vol-FT

The short-end input is a 22-session sequence of log-IV and broadcast normalized-forward return channels; 21 sessions condition the next. The retained architecture uses a causal transformer of width 256, eight layers, four heads and a 200-step cosine diffusion. The fine-tuned variant freezes the EMA base and fits a zero-initialized rank-four output adapter with static-arbitrage penalties. Neither variant is a risk-neutral generator.

Published training settings include learning rate $7 \cdot 10^{-5}$, batch size 256, weight decay 0.01, 1600 warm-up steps, EMA 0.995 and a 3000-epoch initializer. Dated rolling fits warm-start from prior weights; their original normalization is kept fixed because rebasing would change the diffusion-noise law. The newest 20% of training sequences is validation. The April base checkpoint is present but its first next-IV scenario artifact was missing; the audit regenerates 512 CPU paths in eight batches of 64 from the authenticated saved EMA, using batch seeds 20261111–20261118. This new forecast does not pretend to be the missing original 10,000-path sample. The 22 new May FT fits continue the 30 April checkpoint, freeze the original base, and train only the rank-four output adapter for the same one selected epoch per date. All new parent/base hashes and training dates are retained. Forecast and recursive-test completion are shown in Table 11, not inferred from training completion. Recursive diffusion uses all 200 denoising steps each session and the original weekly adapter's context rule: spatially average the generated return channel and rebroadcast that scalar when appending each path's own next state. Per-session checkpoints retain context, generated IVs and RNG state.

Reference: Han, Zhang, Torres, Acero and Xu, Diffusion Models for Dynamic Volatility Surface

Generation and Data-Driven Hedging, arXiv:2609.13402.

J.6 GROMM under P and Q

P and Q share the same 44 dated affine-200 fits and rank-16 dynamics; they are not separately duplicated trained models. The retained native grid uses 200 interior strikes, NMS $(-3, 2)$ and a changing union of reference/quoted expiries, not the proposed new $(-4, 3)$ grid. The Black background annual volatility is 5%. Affine fitting uses 2000 epochs, learning rate 0.01 and DLV-squared bounds 10^{-4}–9. The original fit target is the released SANOS price panel with stored half-spreads, not observed midpoints. Stored training-window fit rates are therefore kept distinct from the direct licensed-quote measurements in this audit.

The retained artifacts were migrated by exact input/geometry reuse between public manifests; their source/target fingerprints and hashes remain in each dated manifest. Some migrated fits have no affine optimizer checkpoint or warm-start date. We do not silently claim they were sequentially retrained. P uses the accepted attainable physical risk-price projection; Q omits that Brownian drift shift. Both use eight Brownian DLV substeps and the original numerical scheme, including its numerical safeguards. An AR-related physical drift estimate is not a global linear VAR transition for the decoded risk-neutral surface; spectral radius is therefore n/a.

The stored native explained-variance ratio of 1.0 on the first fit uses Gaussian-smoothed, centered native log-DLV innovations restricted to the fitted strike core, including the spot projection plus residual factors. It is not 100% coverage of raw full-grid IV changes or an out-of-sample claim. This report instead measures all models directly in the same observable space.

The new observable audit generates 512 antithetic paths (256 shock pairs) per date and measure with seed 20261111 on CPU, using the saved fits without training. It prices the native density tower plus background factor and converts to floating common-grid IVs. Terminal tenors are aged correctly: tomorrow's 1D row comes from today's 2D expiry. This differs from the retained 10,000-path price-return books used for the quoted-book forecast audit. Monte Carlo uncertainty of 512 paths and 44 realized dates must not be conflated. Common random seeds across origins are intentional, not additional independent market observations. Q-versus-P empirical comparisons are explicitly labelled descriptive.

The retained GROMM integrator tames an Euler relative-drift increment x only when $|x| > 0.5$, replacing it with $\mathrm{sgn}(x)[0.5 + \log(1 + |x| - 0.5)]$ before the exponential variance update. This is an implementation stabilization, not the unchanged continuous-time drift restriction and not a proof of exact finite-step martingality. Taming frequency was not retained in the new observable extracts, so no zero-intervention claim is made. Float32 saved density inputs are decoded in float64 and normalized only within the source-precision mass-error tolerance; the diagnostic logs preserve the errors and tolerances.

J.7 Hedge wrappers and non-generative baselines

Greek, parametric and statistical hedge wrappers do not constitute three separately trained surface generators when they reuse the same DYSANOS state and dynamics. Their underlying-law diagnostics are shared. Unhedged, Black delta and Black delta-plus-vega are hedging baselines, not joint stochastic IV-surface models, so covariance, PCA persistence and linear-core spectral radius are not fabricated for them. No hedge objective, risk aversion, transaction-cost setting or optimizer was changed or rerun by this audit.

J.8 Inference-only model retention

At the user's request, the diffusion models now have an EMA-only inference archive with exact content-addressed backbone sharing. 88 dated exports have been checked tensor by tensor and through two

seeded complete 200-step CPU sampling paths against their original checkpoints, with zero differences. Geometry, versions, normalization, dated adapters and historical source hashes are retained. 88 redundant restart checkpoints (4.430 GiB gross) have been retired; the inference archive occupies 0.756 GiB. This preserves inference, not exact optimizer/RNG restart. Original fitted laws are unchanged; inference operability is not a calibration-quality pass. Other models, historical contexts, retained scenarios and results have not been deleted by this step.

K Reproducibility and missing evidence

Licensed data and fitted parameters are unchanged. Large models, retained scenario arrays and licensed quote rows remain external; only aggregates, diagnostics and checksummed manifests are suitable for Git. Retired checkpoint identities resolve through `INFERENCE_MODELS.json`; `model_audit/inference.py` loads the verified inference replacements without optimizer or training data.

The report is generated by `quantdata/hedging/model_audit/report.py`; `run.py`, `persistence.py`, `gromm.py` and `quotes.py` reproduce its stages. Interpreter: `/home/hans/vqdata/bin/python`. Jobs set explicit `QUANTDATA_ROOT` and `PYTHONPATH`. Neural recursive paths use a managed GPU; other audit stages run on CPU. Disjoint origin batches preserve each origin's seed, saved context and RNG state. No diffusion denoising steps are dropped.

The JSON daily files retain failure reasons, no-data exclusions, quantile forecasts, full conditional covariance and source diagnostics. CSVs provide coverage, daily/monthly/rolling errors, per-expiry/NMS log-IV RMSE, tail tests and linear-core stability. Raw modeled means are retained: this is a model-law audit, not the demeaned hedge-optimization experiment. The 44-day report cannot establish 63-day observed ACF or reliable 1% tail calibration. Missing diagnostics remain n/a.